

COURSE CODE/TITLE:	CPE 212 - Automating Server Management Server	SECTION:	CPE31S2
DATE PERFORMED:	09/07/2026	DATE SUBMITTED:	09/07/2026
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ACTIVITY NO. 1.1 Hands-on Activity - Creating Virtual Machines

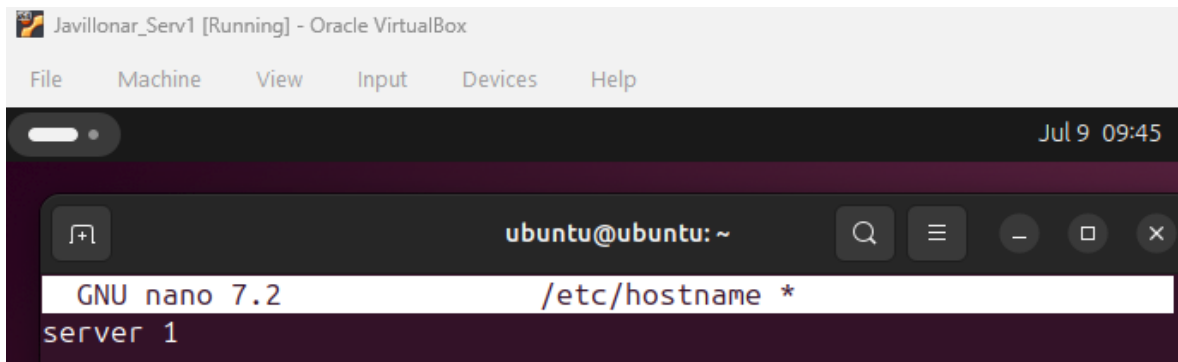
1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

1. Change the hostname using the command *sudo nano /etc/hostname*

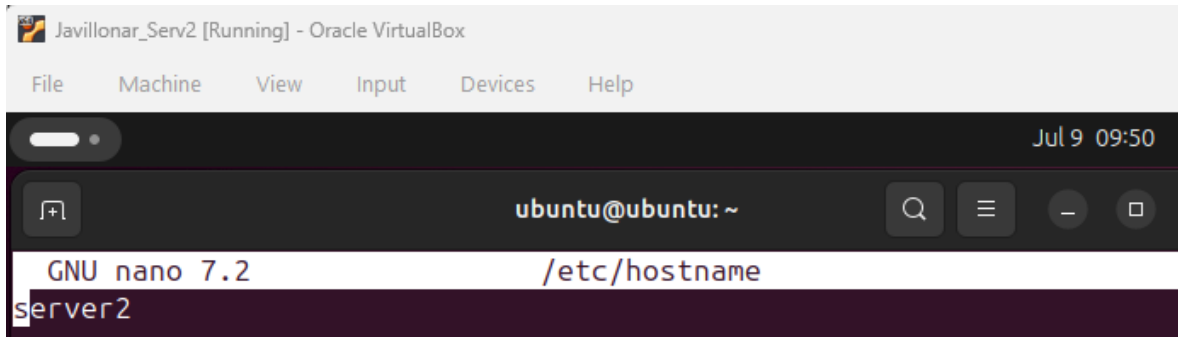
1.1 Use server1 for Server 1



```

Javillonar_Serv1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:45
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hostname *
server 1
  
```

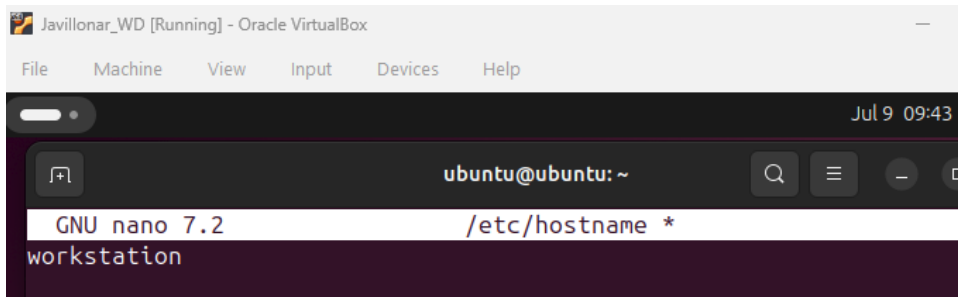
1.2 Use server2 for Server 2



```

Javillonar_Serv2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:50
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hostname
server2
  
```

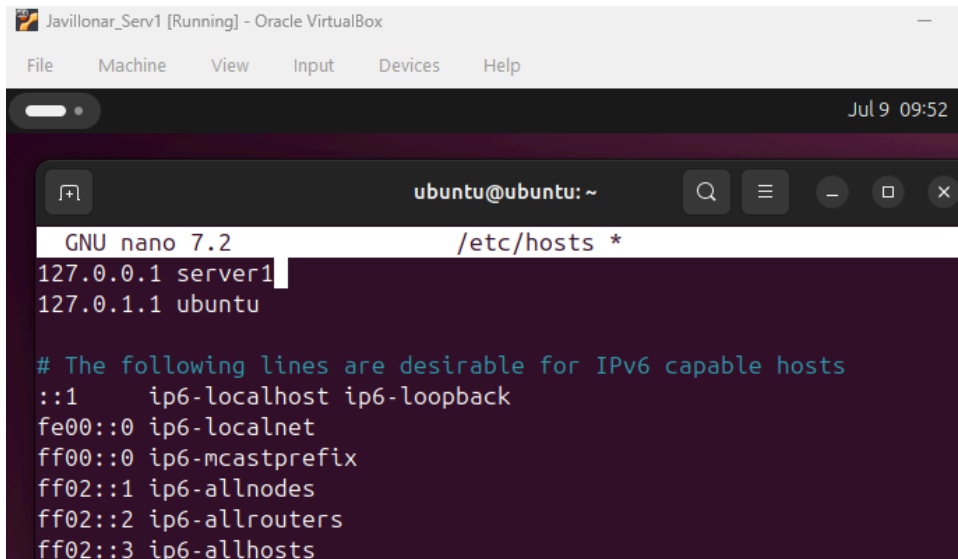
1.3 Use workstation for the Local Machine



```
Javillonar_WD [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:43
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hostname *
workstation
```

2. Edit the hosts using the command `sudo nano /etc/hosts`. Edit the second line.

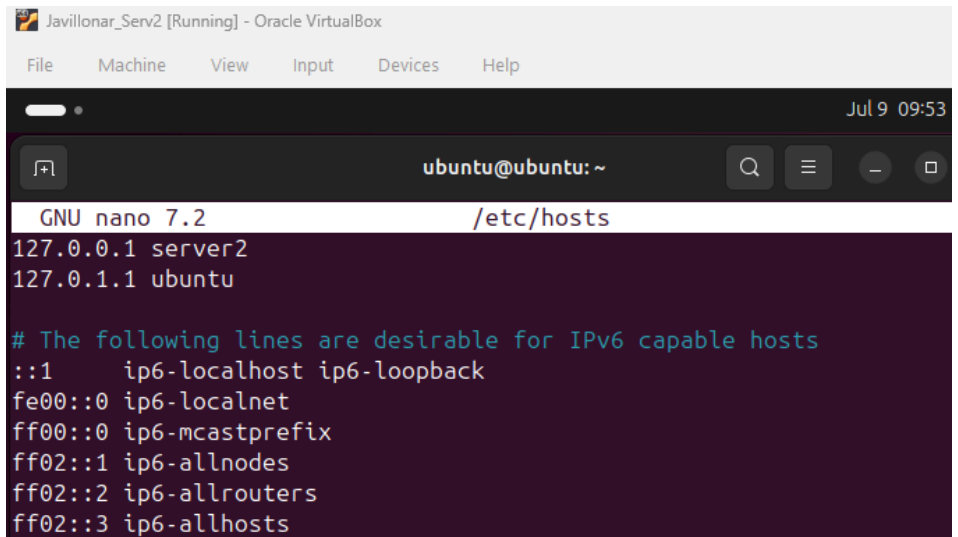
2.1 Type 127.0.0.1 server 1 for Server 1



```
Javillonar_Serv1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:52
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hosts *
127.0.0.1 server1
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

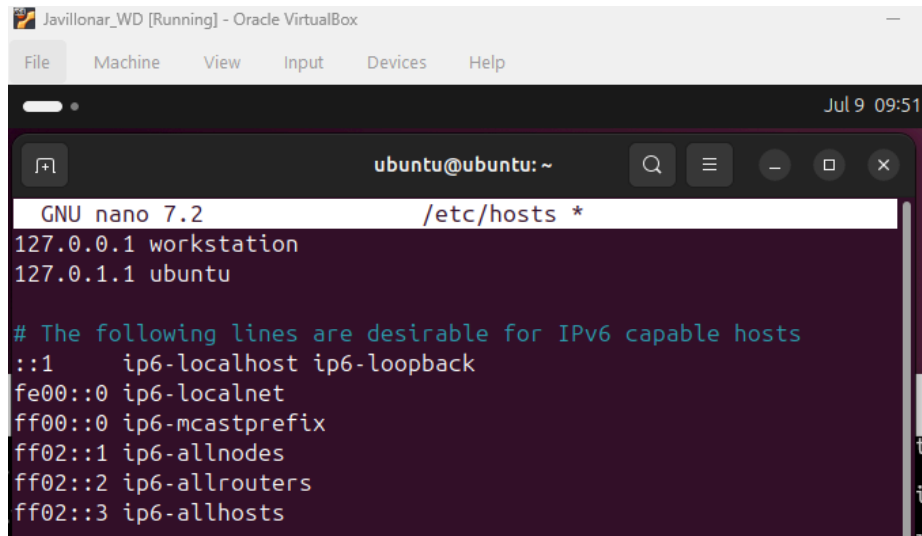
2.2 Type 127.0.0.1 server 2 for Server 2



```
Javillonar_Serv2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:53
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hosts
127.0.0.1 server2
127.0.1.1 ubuntu

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

2.3 Type 127.0.0.1 workstation for the Local Machine



```
Javillonar_WD [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 9 09:51
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/hosts *
127.0.0.1 workstation
127.0.1.1 ubuntu

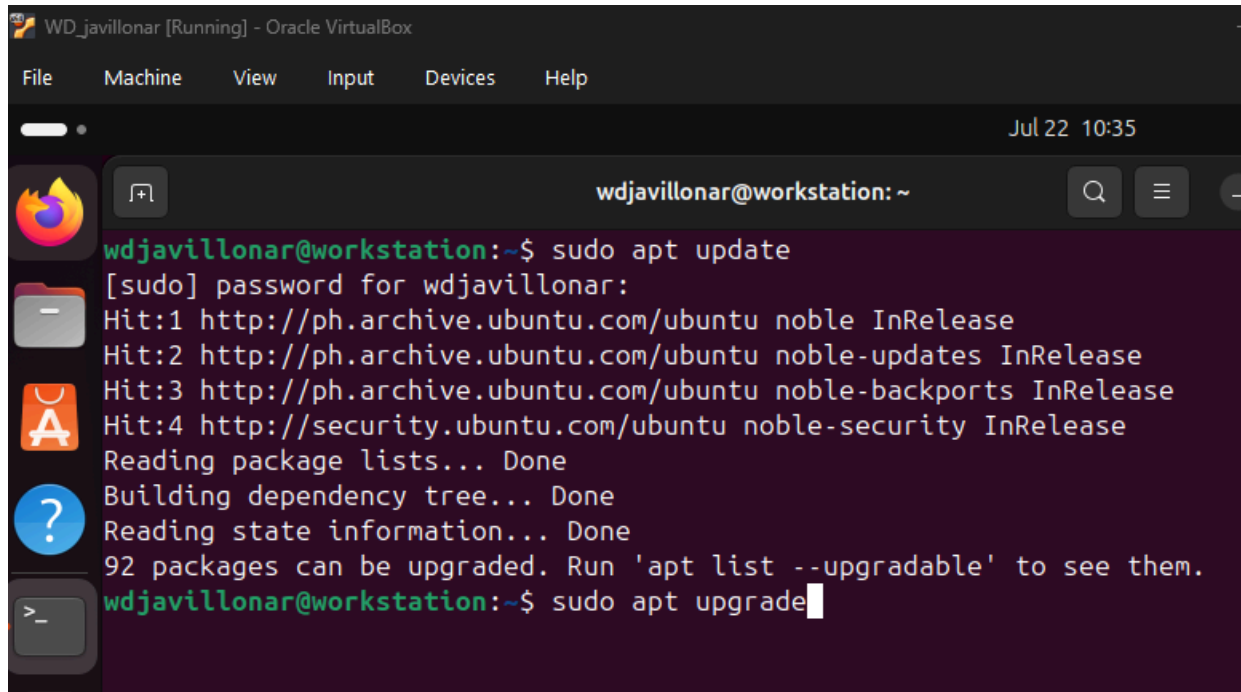
# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
ff02::3  ip6-allhosts
```

Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.

Local Machine

(sudo apt update)



```
WD_javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:35
wdjavillonar@workstation: ~
wdjavillonar@workstation:~$ sudo apt update
[sudo] password for wdjavillonar:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
wdjavillonar@workstation:~$ sudo apt upgrade
```

(sudo apt upgrade)

```
WD_javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:35
wdjavillonar@workstation: ~
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
wdjavillonar@workstation:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  libfwupd3
The following packages will be upgraded:
  alsa-ucm-conf apparmor appport appport-core-dump-handler appport-gtk cloud-init
  coreutils fwupd gdb gdm3 gir1.2-gdm-1.0 gir1.2-gtk-4.0 gir1.2-mutter-14
  gir1.2-nm-1.0 gnome-control-center gnome-control-center-data
  gnome-control-center-faces gnome-keyring gnome-keyring-pkcs11 gnome-shell
  gnome-shell-common iproute2 libapparmor1 libfprint-2-2 libfprint-2-tod1
  libfwupd2 libgdm1 libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64
  libgtk-4-1 libgtk-4-bin libgtk-4-common libgtk-4-media-gstreamer
```

Server 1

(sudo apt update)

```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:36
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo apt update
[sudo] password for serv1javillonar:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

(sudo apt upgrade)

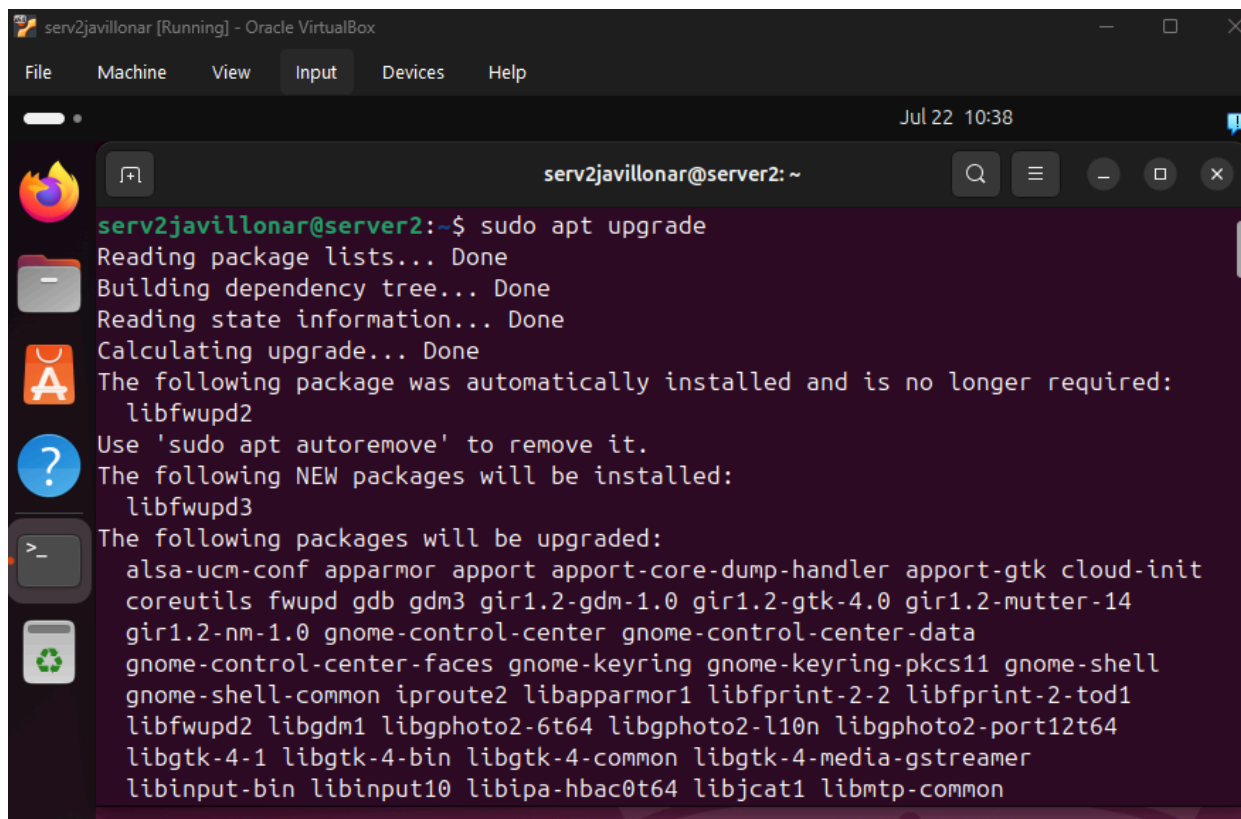
```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:36
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  libfwupd3
The following packages will be upgraded:
  alsa-ucm-conf apparmor apport apport-core-dump-handler apport-gtk cloud-init
  coreutils fwupd gdb gdm3 gir1.2-gdm-1.0 gir1.2-gtk-4.0 gir1.2-mutter-14
  gir1.2-nm-1.0 gnome-control-center gnome-control-center-data
  gnome-control-center-faces gnome-keyring gnome-keyring-pkcs11 gnome-shell
  gnome-shell-common iproute2 libapparmor1 libfprint-2-2 libfprint-2-tod1
  libfwupd2 libgdm1 libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64
  libgtk-4-1 libgtk-4-bin libgtk-4-common libgtk-4-media-gstreamer
  libinput-bin libinput10 libipa-hbac0t64 libjcat1 libmtp-common
```

Server 2

(sudo apt update)

```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:36
serv2javillonar@server2: ~
serv2javillonar@server2:~$ sudo apt update
[sudo] password for serv2javillonar:
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
92 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

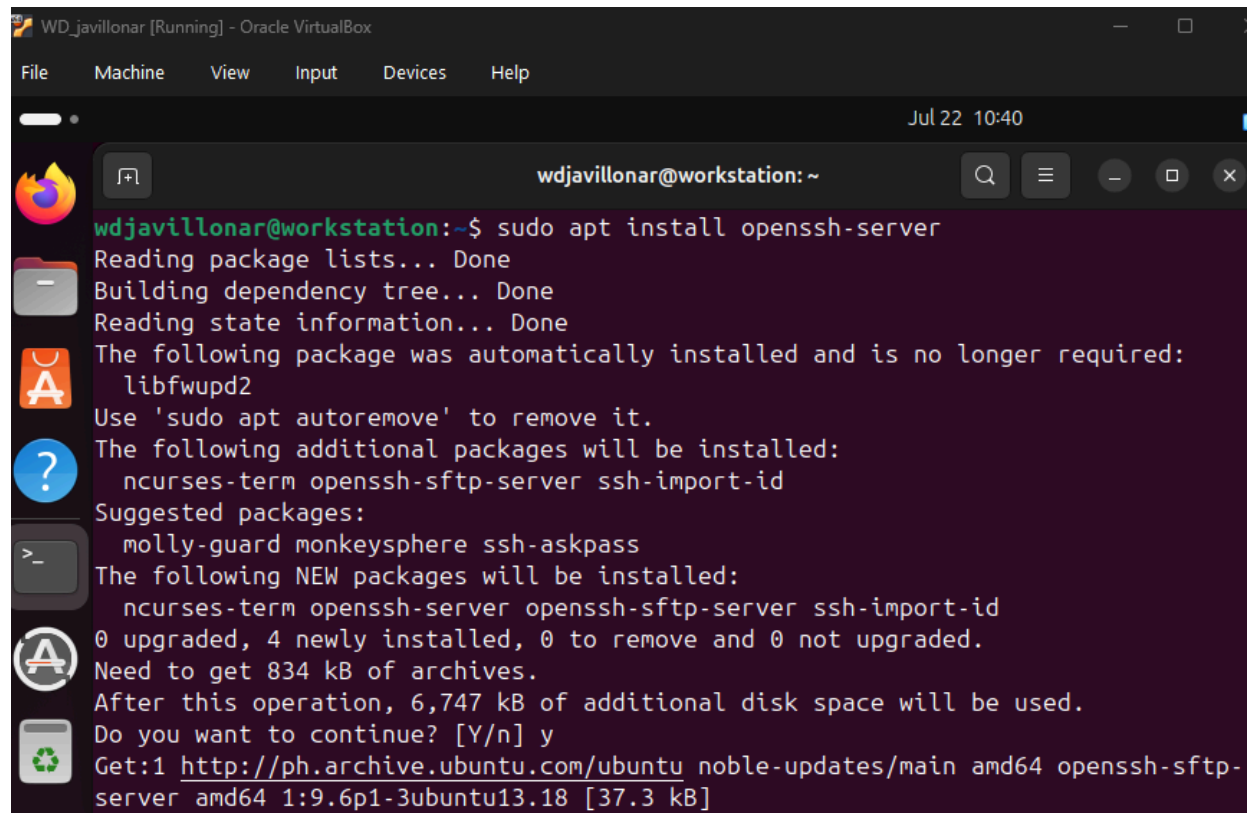
(sudo apt upgrade)



```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:38
serv2javillonar@server2: ~
serv2javillonar@server2:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  libfwupd3
The following packages will be upgraded:
  alsa-ucm-conf apparmor apport apport-core-dump-handler apport-gtk cloud-init
  coreutils fwupd gdb gdm3 gir1.2-gdm-1.0 gir1.2-gtk-4.0 gir1.2-mutter-14
  gir1.2-nm-1.0 gnome-control-center gnome-control-center-data
  gnome-control-center-faces gnome-keyring gnome-keyring-pkcs11 gnome-shell
  gnome-shell-common iproute2 libapparmor1 libfprint-2-2 libfprint-2-tod1
  libfwupd2 libgdm1 libgphoto2-6t64 libgphoto2-l10n libgphoto2-port12t64
  libgtk-4-1 libgtk-4-bin libgtk-4-common libgtk-4-media-gstreamer
  libinput-bin libinput10 libipa-hbac0t64 libjcat1 libmtp-common
```

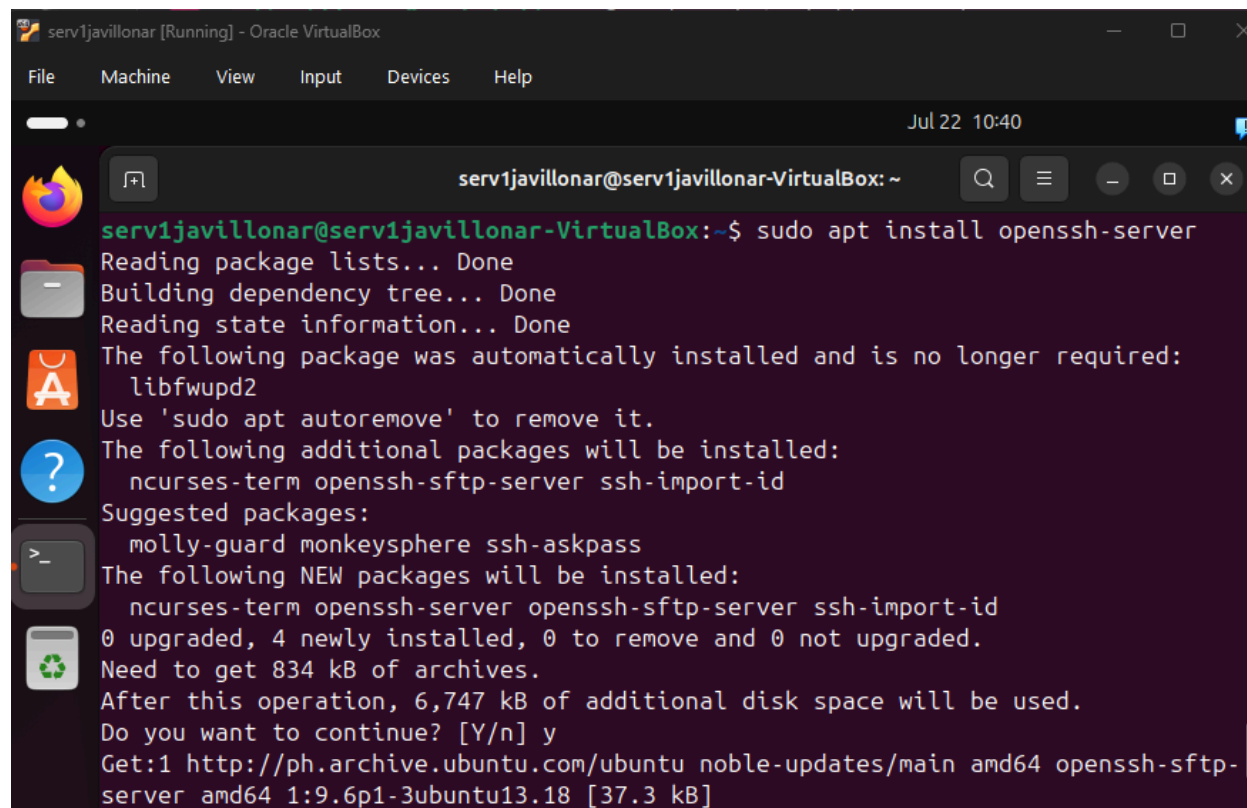
2. Install the SSH server using the command *sudo apt install openssh-server*.

Local Machine



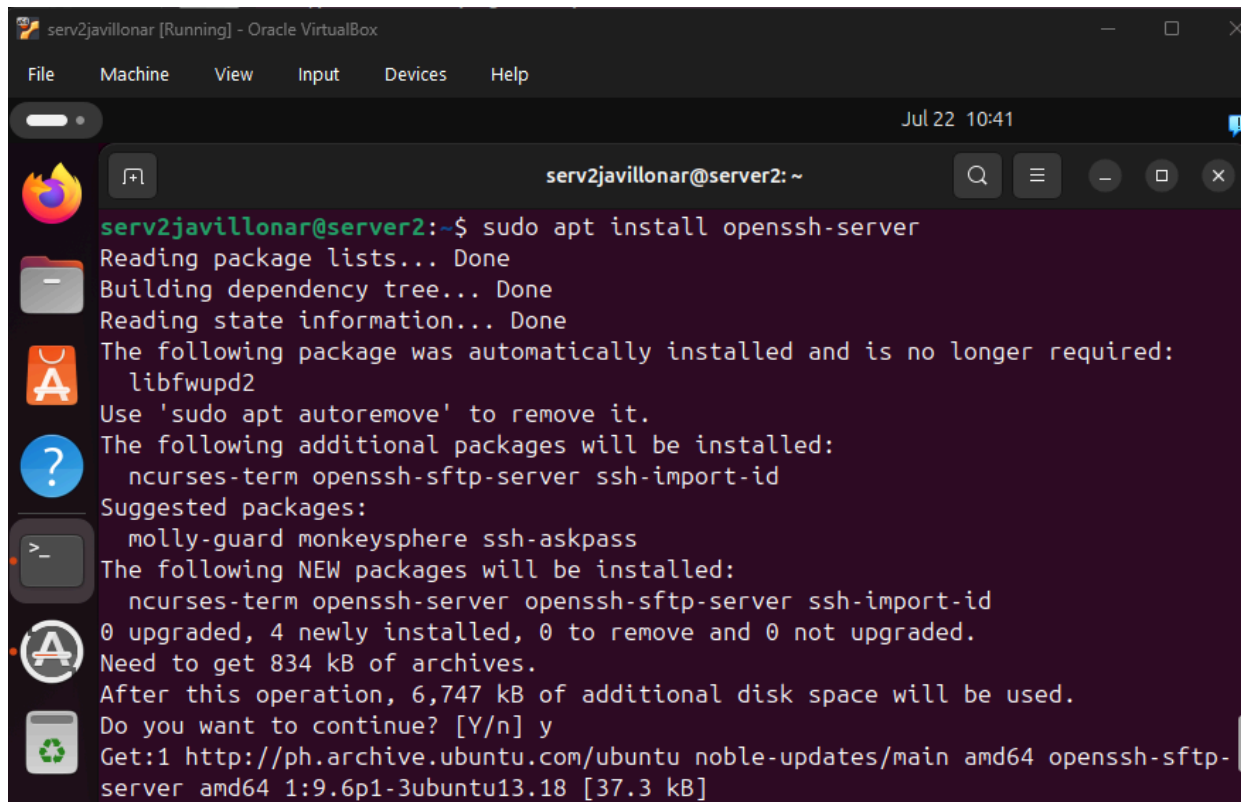
```
WD_javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:40
wdjavillonar@workstation: ~
wdjavillonar@workstation:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 834 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-
server amd64 1:9.6p1-3ubuntu13.18 [37.3 kB]
```

Server 1



```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:40
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 834 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-
server amd64 1:9.6p1-3ubuntu13.18 [37.3 kB]
```

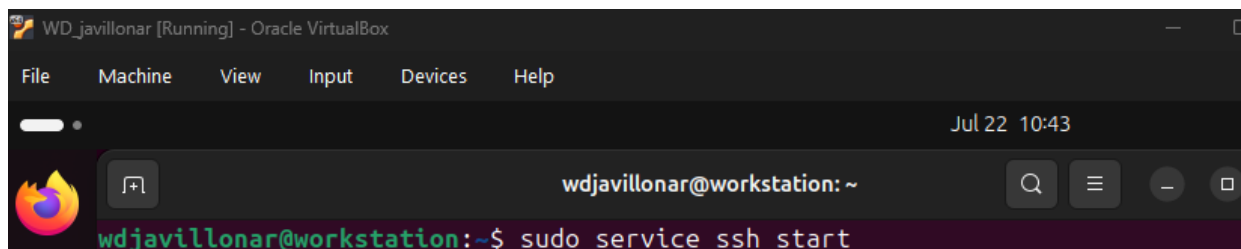
Server 2



```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:41
serv2javillonar@server2: ~
serv2javillonar@server2:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 834 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-
server amd64 1:9.6p1-3ubuntu13.18 [37.3 kB]
```

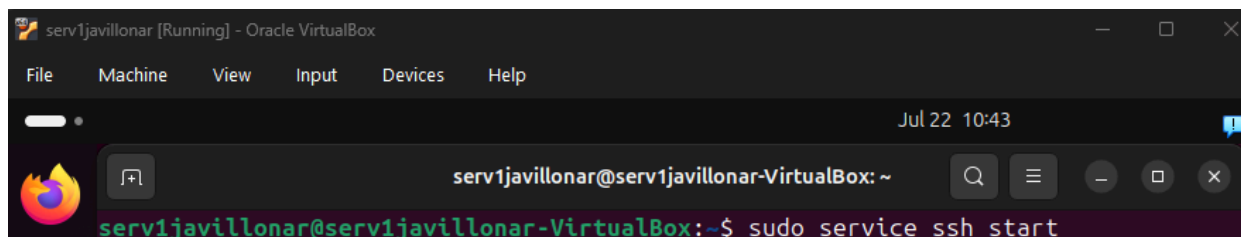
3. Verify if the SSH service has started by issuing the following commands:
 - 3.1 *sudo service ssh start*

Local Machine



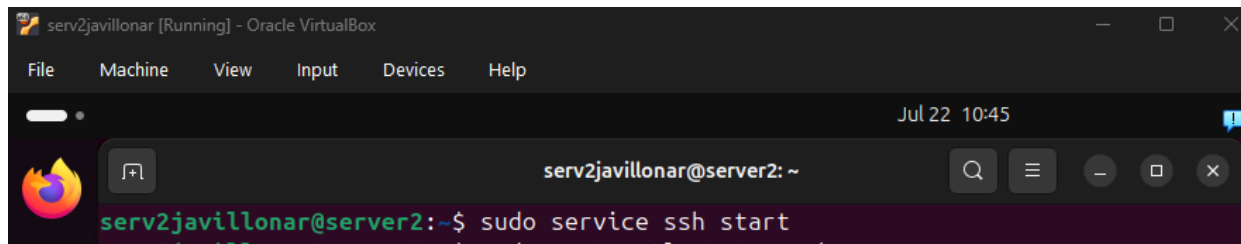
```
WD_javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:43
wdjavillonar@workstation: ~
wdjavillonar@workstation:~$ sudo service ssh start
```

Server 1



```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:43
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo service ssh start
```

Server 2



3.2 *sudo systemctl status ssh*

Local Machine

```
wdjavillonar@workstation:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: ena>
   Active: active (running) since Wed 2026-07-22 10:41:21 PST; 43s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 17414 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS>
   Main PID: 17416 (sshd)
     Tasks: 1 (limit: 3465)
    Memory: 1.2M (peak: 1.7M)
       CPU: 21ms
    CGroup: /system.slice/ssh.service
           └─17416 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 22 10:41:21 workstation systemd[1]: Starting ssh.service - OpenBSD Secure S>
Jul 22 10:41:21 workstation sshd[17416]: Server listening on 0.0.0.0 port 22.
Jul 22 10:41:21 workstation sshd[17416]: Server listening on :: port 22.
```

Server 1

```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:46
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo service ssh start
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2026-07-22 10:41:36 PST; 1min 23s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 17920 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 17921 (sshd)
     Tasks: 1 (limit: 3465)
    Memory: 1.2M (peak: 1.5M)
       CPU: 21ms
    CGroup: /system.slice/ssh.service
            └─17921 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 22 10:41:36 serv1javillonar-VirtualBox systemd[1]: Starting ssh.service - 0>
Jul 22 10:41:36 serv1javillonar-VirtualBox sshd[17921]: Server listening on 0.0.0.0 port 22.
Jul 22 10:41:36 serv1javillonar-VirtualBox sshd[17921]: Server listening on :: port 22.
```

Server 2

```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:45
serv2javillonar@server2: ~
serv2javillonar@server2:~$ sudo service ssh start
serv2javillonar@server2:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2026-07-22 10:41:45 PST; 2min 23s ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Process: 17663 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 17665 (sshd)
     Tasks: 1 (limit: 3465)
    Memory: 1.2M (peak: 1.6M)
       CPU: 20ms
    CGroup: /system.slice/ssh.service
            └─17665 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 22 10:41:45 server2 systemd[1]: Starting ssh.service - OpenBSD Secure Shell
Jul 22 10:41:45 server2 sshd[17665]: Server listening on 0.0.0.0 port 22.
Jul 22 10:41:45 server2 sshd[17665]: Server listening on :: port 22.
```

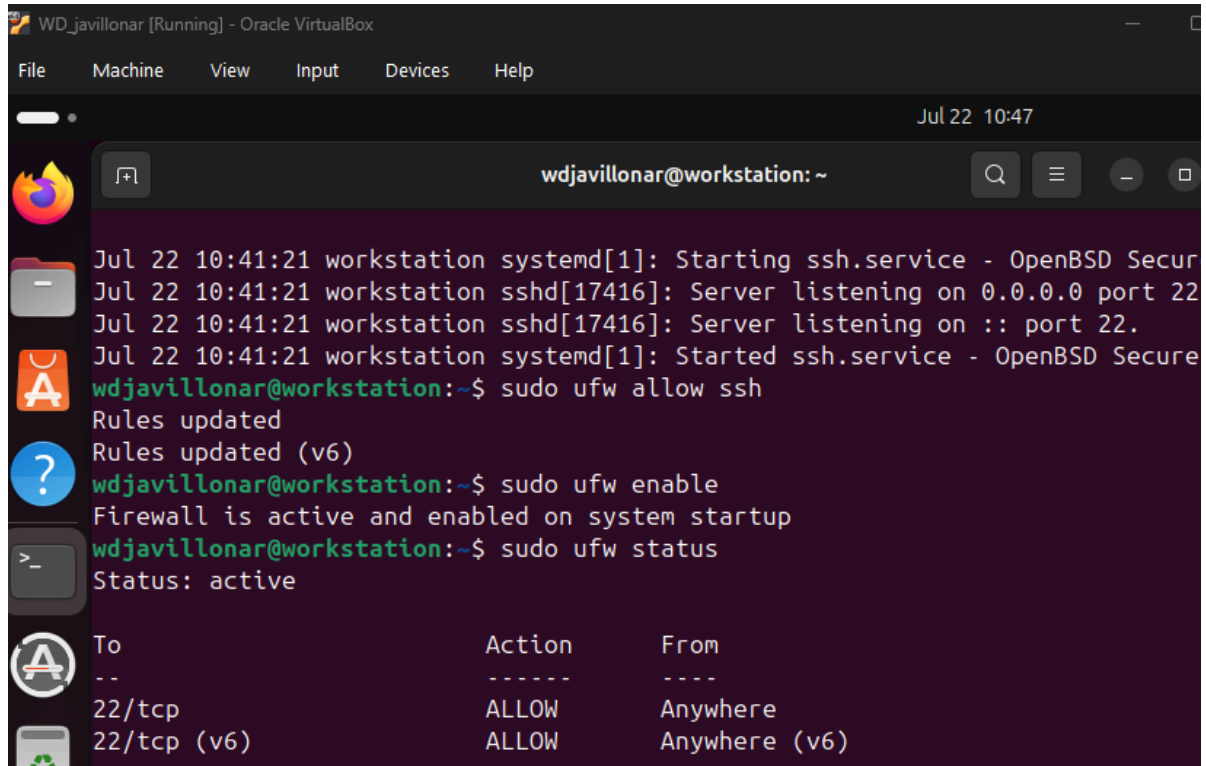
4. Configure the firewall to all port 22 by issuing the following commands:

4.1 *sudo ufw allow ssh*

4.2 *sudo ufw enable*

4.3 *sudo ufw status*

Local Machine



```
WD_javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:47
wdjavillonar@workstation: ~
Jul 22 10:41:21 workstation systemd[1]: Starting ssh.service - OpenBSD Secur
Jul 22 10:41:21 workstation sshd[17416]: Server listening on 0.0.0.0 port 22
Jul 22 10:41:21 workstation sshd[17416]: Server listening on :: port 22.
Jul 22 10:41:21 workstation systemd[1]: Started ssh.service - OpenBSD Secure
wdjavillonar@workstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
wdjavillonar@workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
wdjavillonar@workstation:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

Server 1

```
serv1javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:48
serv1javillonar@serv1javillonar-VirtualBox: ~
Jul 22 10:41:36 serv1javillonar-VirtualBox systemd[1]: Starting ssh.service - OpenSSH
Jul 22 10:41:36 serv1javillonar-VirtualBox sshd[17921]: Server listening on 0.0.0.0
Jul 22 10:41:36 serv1javillonar-VirtualBox sshd[17921]: Server listening on ::
Jul 22 10:41:36 serv1javillonar-VirtualBox systemd[1]: Started ssh.service - OpenSSH
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
serv1javillonar@serv1javillonar-VirtualBox:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
```

Server 2

```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:49
serv2javillonar@server2: ~
Jul 22 10:41:45 server2 systemd[1]: Started ssh.service - OpenSSH Secure Shell
serv2javillonar@server2:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
serv2javillonar@server2:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Jul 22 10:49
serv2javillonar@server2: ~
status verbose          show verbose firewall status
show ARG                show firewall report
version                 display version information

Application profile commands:
app list                list application profiles
app info PROFILE        show information on PROFILE
app update PROFILE      update PROFILE
app default ARG         set default application policy

serv2javillonar@server2:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

serv2javillonar@server2:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

- Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
 - Server 1 IP address: 192.168.56.101
 - Server 2 IP address: 192.168.56.102
 - Server 3 IP address: 192.168.56.103 (wd)

```
wdjavillonar@workstation: ~
wdjavillonar@workstation:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:70ce:4758:54ac:c829 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:fe8c:9d2e prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe8c:9d2e prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:8c:9d:2e txqueuelen 1000 (Ethernet)
    RX packets 72 bytes 9991 (9.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 165 bytes 21215 (21.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::cb4d:59a6:d02c:1446 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b7:1f:74 txqueuelen 1000 (Ethernet)
    RX packets 183 bytes 29351 (29.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 62 bytes 8055 (8.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

serv1javillonar [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Jul 22 13:38

```
serv1javillonar@serv1javillonar-VirtualBox: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:feb3:66ef prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:feb3:66ef prefixlen 64 scopeid 0x0<global>
    inet6 fd17:625c:f037:2:3029:39e2:6fcf:3b1c prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:b3:66:ef txqueuelen 1000 (Ethernet)
    RX packets 66 bytes 9318 (9.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 165 bytes 22005 (22.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a294:7442:9e6e:9902 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a5:4c:f5 txqueuelen 1000 (Ethernet)
    RX packets 39 bytes 7161 (7.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 64 bytes 8484 (8.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
```

```
serv2javillonar [Running] - Oracle VirtualBox
File Machine View Input Devices Help
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serv2javillonar@server2: ~
serv2javillonar@server2:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:10ca:cb32:3f0d:cde8 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::a00:27ff:feb3:4967 prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:feb3:4967 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:b3:49:67 txqueuelen 1000 (Ethernet)
    RX packets 101 bytes 20909 (20.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 190 bytes 24130 (24.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::b747:99c6:f979:3fe0 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:d4:20:90 txqueuelen 1000 (Ethernet)
    RX packets 157 bytes 24740 (24.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 65 bytes 8368 (8.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☒ Successful ☐ Not Successful

```
wdjavillonar@workstation:~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.04 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.558 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.575 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.757 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.01 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.785 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.551 ms
^C
--- 192.168.56.101 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6072ms
rtt min/avg/max/mdev = 0.551/0.753/1.035/0.192 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: ☒ Successful ☐ Not Successful

```
wdjavillonar@workstation:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.640 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.601 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.652 ms
^Z
[1]+  Stopped                  ping 192.168.56.102
```

2.3 Connectivity test for Server 1 to Server 2: ☒ Successful ☐ Not Successful.

```
serv1javillonar@serv1javillonar-VirtualBox:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.72 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.635 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.711 ms
^Z
[1]+  Stopped                  ping 192.168.56.102
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

- 1.1 `ssh username@ip_address_server1` for example, `ssh jvtaylor@192.168.56.120`
- 1.2 Enter the password for server 1 when prompted



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```
WD_javillonar [Running] - Oracle VM VirtualBox
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wdjavillonar@workstation: ~
serv1javillonar@serv1javillonar-VirtualBox:~$ exit
logout
Connection to 192.168.56.101 closed.
wdjavillonar@workstation:~$ ssh serv1javillonar@192.168.56.101
serv1javillonar@192.168.56.101's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jul 22 13:49:26 2026 from 192.168.56.103
serv1javillonar@serv1javillonar-VirtualBox:~$
logout
Connection to 192.168.56.101 closed.
wdjavillonar@workstation:~$
```

```

wdjavillonar@workstation: ~
wdjavillonar@workstation:~$ ssh serv2javillonar@192.168.56.102
serv2javillonar@192.168.56.102's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
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individual files in /usr/share/doc/*/copyright.

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applicable law.

serv2javillonar@server2:~$
logout
Connection to 192.168.56.102 closed.

```

- 1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylor@server1*
2. Logout of Server 1 by issuing the command *control + D*.
3. Do the same for Server 2.
4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts*. Below all texts type the following:
 - 4.1 *IP_address server 1* (provide the ip address of server 1 followed by the hostname)
 - 4.2 *IP_address server 2* (provide the ip address of server 2 followed by the hostname)

```
wdjavillonar@workstation: ~  
GNU nano 7.2 /etc/hosts  
127.0.0.1 localhost  
127.0.1.1 workstation  
  
192.168.56.101 server1  
192.168.56.102 server2  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters
```

- 4.3 Save the file and exit.
5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do `ssh jvtaylor@server1`. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

```
wdjavillonar@workstation:~$ ssh serv1javillonar@192.168.56.101
serv1javillonar@192.168.56.101's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jul 22 13:51:33 2026 from 192.168.56.103
serv1javillonar@serv1javillonar-VirtualBox:~$
logout
Connection to 192.168.56.101 closed.
```

```
wdjavillonar@workstation:~$ ssh serv2javillonar@192.168.56.102
serv2javillonar@192.168.56.102's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

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0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jul 22 13:52:46 2026 from 192.168.56.103
serv2javillonar@server2:~$
logout
Connection to 192.168.56.102 closed.
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
 - We are able to use the hostname instead of the IP address in SSH commands because the hostname is linked to the machine's IP address through the `/etc/hosts` file. When we type the hostname, the system automatically finds the correct IP address and connects to that machine. This makes it easier to remember and use compared to typing the full IP address every time.
2. How secured is SSH?
 - SSH or Secure Shell is very secure since it can be used as the global standard for a secure remote administration. SSH is also known that it has a different options for user authentication such as public or private key and can customize the public key inside a server just to authorize access and give anyone who has an access or copy of the private key access to the server. Lastly, SSH offers a secure encryption for data integrity because both of the SSH client and server have established a connection, the transmitted data is encrypted based on the parameters agreed upon during the setup phase. In this negotiation process, both the client and the server decide on the symmetric encryption algorithm to implement and create the key that will facilitate the encryption

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

QuillBot
Gemini

Purpose of Use:

I used Quillbot for grammar correction and paraphraser and Gemini for searching informations.

Extent of Use:

I confirm that AI didn't replace any of my original ideas in accomplishing the today's activity since, I just used it as an assistance for enlightening my ideas and corrected my grammars.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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